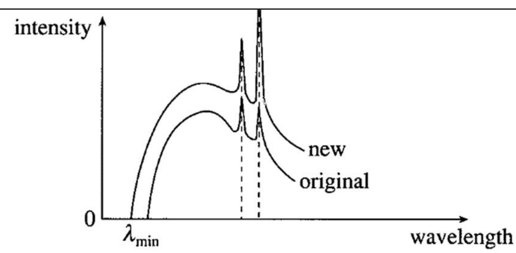
7(a)	Each metal has a characteristic minimum amount of energy, needed to liberate an electron from its surface. This characteristic minimum amount of energy needed is called the work function energy, Φ of the metal.	
7(b)(i)	$E = hf$ $= \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})(3\times10^{8})}{540\times10^{-9}}$ $= 3.68\times10^{-19}$ J	

7(b)(ii)	Intensity = $\frac{power}{area}$ $= \frac{Energy}{time \times area} = \frac{nhf}{tA}$ $= 150 = \frac{n}{t}(\frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{540 \times 10^{-9} \times 14 \times 10^{-6}})$ $\frac{n}{t} = 5.7 \times 10^{15}$ photons per unit time	
7(b)(iii)	$E = hf$ $= \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{540 \times 10^{-9}}$ $= 3.68 \times 10^{-19}$ J $= 2.3$ eV $hf = \phi + KE_{\max}$ $KE_{\max} = 2.3 - 1.9 = 0.4$ eV $= 6.43 \times 10^{-20}$ J	
7(c)(i)	As the bombarding electrons are <u>slowed down suddenly upon hitting the target,</u> X-rays are emitted and forms the continuous part of the spectrum. An electron may lose all or part of its energy in such a collision. The continuous X-ray spectrum is explained as being due to such <u>***bremsstrahlung***</u> collisions in which <u>varying amounts of energy are lost by the electrons.</u> An electron is <u>stopped by a single (instead of multiple) impact and all its kinetic energy is used to produce one photon.</u> This X-ray photon will possess the maximum possible energy and correspondingly, the shortest wavelength $\lambda_{\min}$.	
7(c)(ii)	 When the potential difference increases, $\lambda_{\min}$ decreases, since $eV_{AC} = \frac{hc}{\lambda_{\min}}$.	

	Characteristic X-rays remain at the same λ since it is characteristic of the target. For the same atoms, the atomic transitions involved remain unchanged and hence the wavelengths of the emitted photons are unchanged.	
8(a)	There are 3 positively charged particles as there are 3 lines	